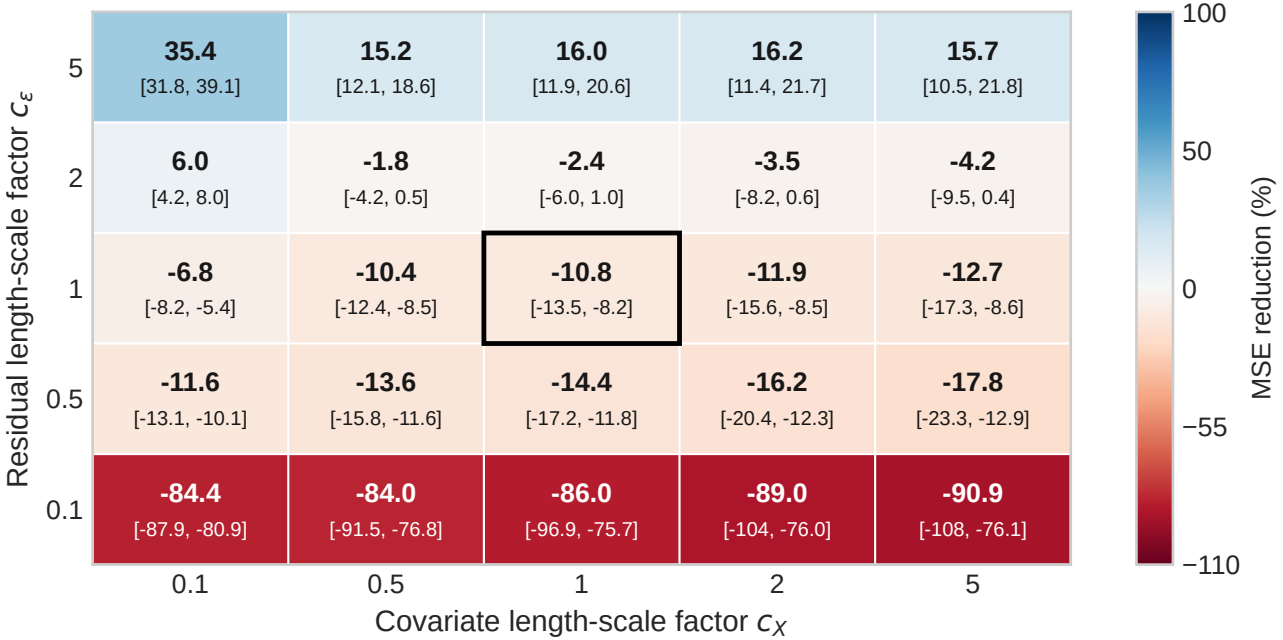
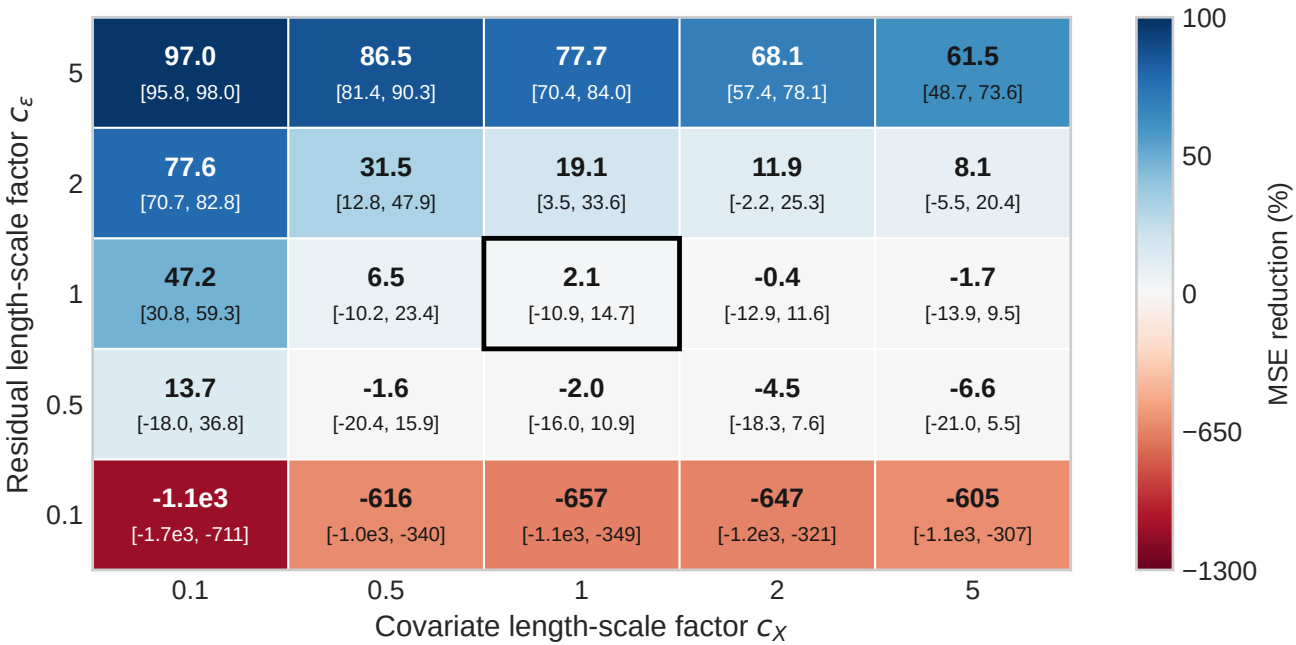
$N = 500,\ s_m = 1,\ s_\sigma = 0.5,\ \rho = 0.3$

Operator MSE: $\mathbb{E}\|\widehat{\Psi} - \Psi\|^2_{\mathrm{HS}}$



HSIC U-statistic MSE: $\mathbb{E}(U - \theta)^2$



At each bandwidth pair: debiased MSE compared with plug-in MSE.

Blue favors debiasing; red favors plug-in. Brackets: 95% Monte Carlo intervals.